



Understanding Complexity and Interactivity of Brain Patterns

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Overview

We hypothesize that human thought and complex cognition are supported by higher-order dynamic interactions between distributed brain regions.

Developed a computationally efficient framework (Timecorr) to measure higher-order correlations in multidimensional time-series data.

Applied this approach to fMRI data collected as participants listened to a 10-minute story under varying coherence conditions (intact, paragraph-scrambled, word-scrambled, rest).

Higher-order correlations reliably distinguished between narrative conditions, capturing integrative dynamics beyond pairwise connectivity.

Introduced scalable, modular improvements to Timecorr, including automated parameter sweeps, expanded kernel functions, and optimized higher-order correlation computation, making the toolbox faster, more flexible, and more reproducible.

Background

Interested in uncovering the neural activity patterns and network dynamics that underlie higher-order cognition and complex thought.

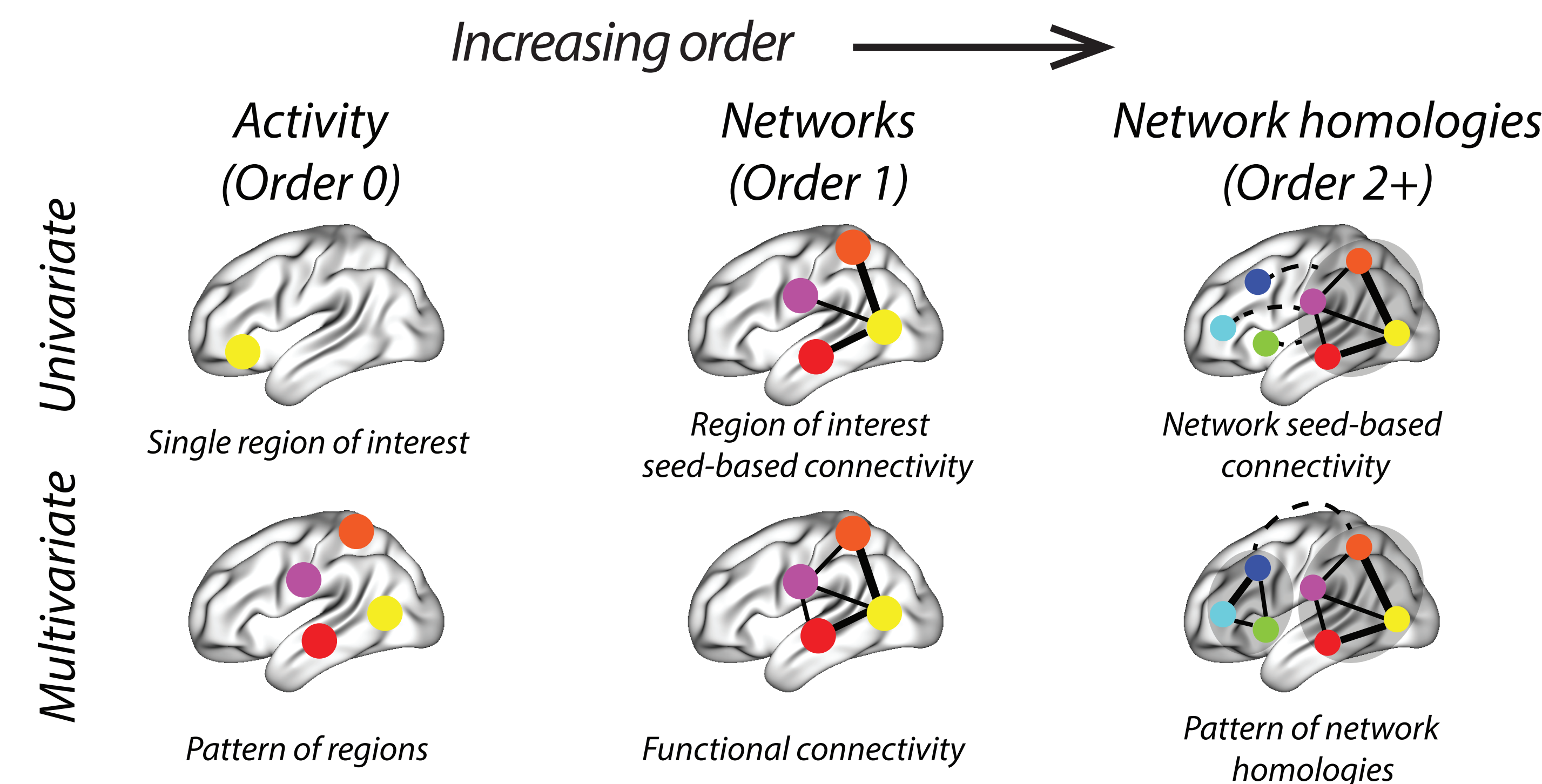


Fig. 1 Neural patterns. Univariate analyses characterize the activities of individual units (e.g., nodes, small networks, hierarchies of networks, etc.), whereas multivariate analyses characterize the patterns of activities across units. Order 0 patterns involve individual nodes; order 1 patterns involve node-node interactions; order 2 (and higher) patterns relate to interactions between homologous networks. Each of these patterns may be static (e.g., averaging over time) or dynamic.

Can dynamic higher-order interactions help decode mental states?

If so, what order of interactions are most reflective of high-level cognitive processes?

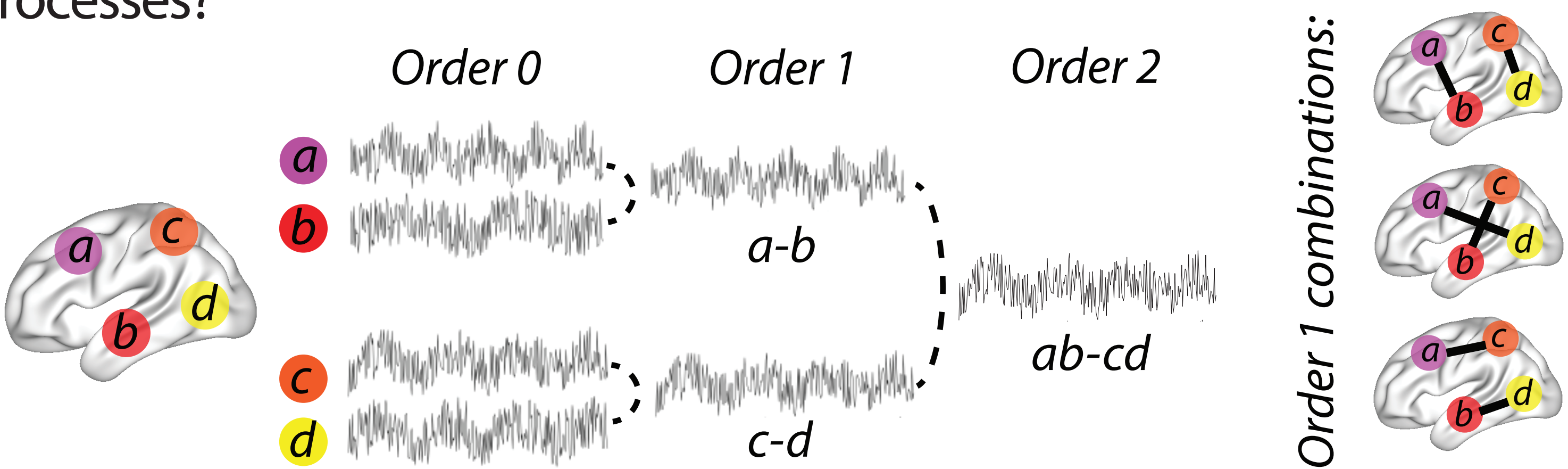


Fig. 2 Example of dynamic higher-order correlations. Raw activity (order 0) for 4 locations (a, b, c, and d) are shown. Dynamic correlations (order 1) show how a and b as well as how c and d are related over time. Dynamic order 2 correlations reflect the correlation between those pairs of order 1 correlations. Also shown are the other possible order 1 combinations.

Approach

Dynamic Correlations

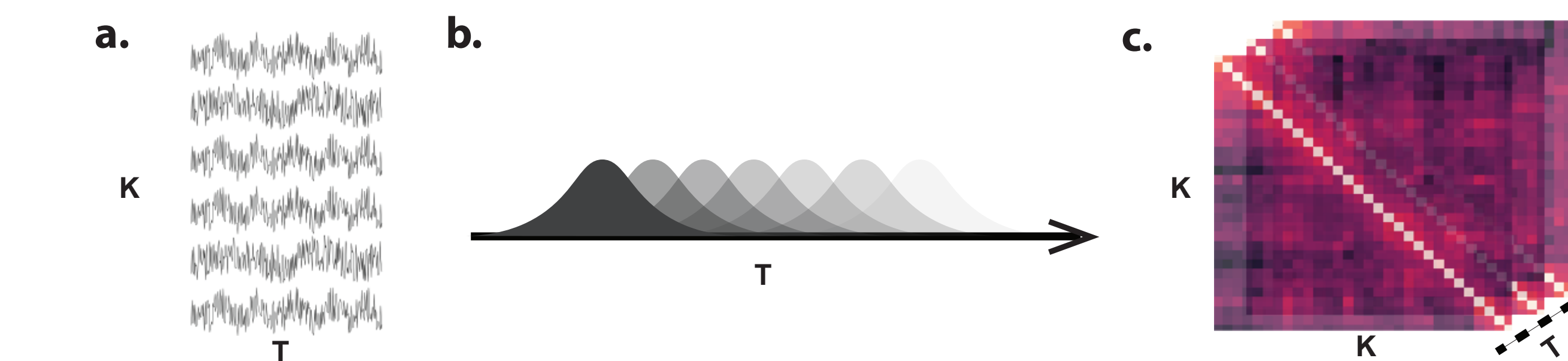


Fig 3. Computer dynamic correlations. a. Timepoints (T) by features (K) data. b. Compute the time-varying correlations by incorporating a weight function. A gaussian weight is shown here, but other options include: Laplace, mexican hat, and delta. c. Returns dynamic

Higher-Order Correlations

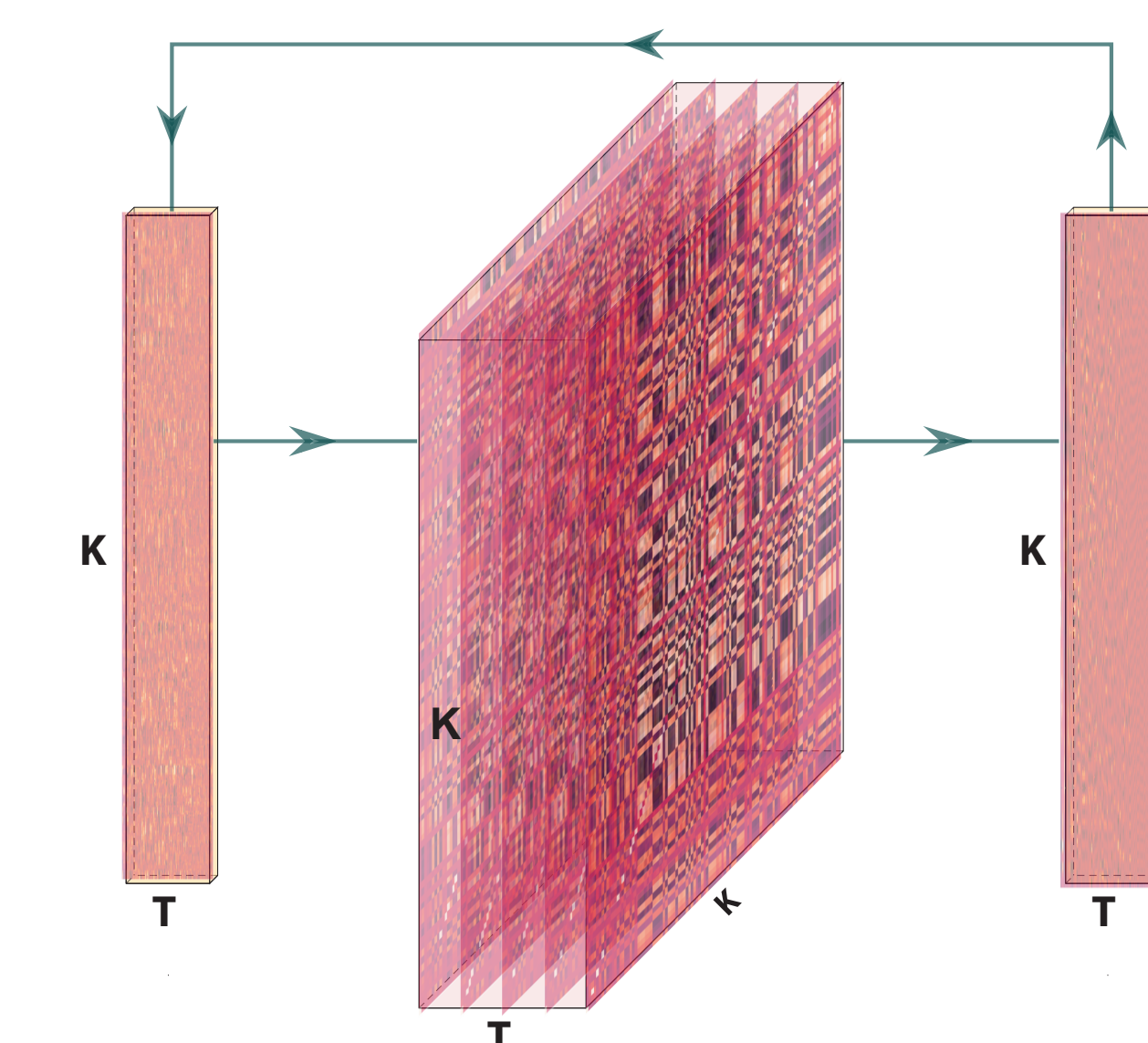


Fig 4. Estimate dynamic high-order correlations. a. Given timeseries data (TxK) we compute the dynamic correlations (TxKxK). To efficiently compute with complex neural patterns, we then approximate these patterns (TxK) using summary measures. b. Dimensionality reduction algorithms project the patterns onto lower-dimensional spaces whose dimensions reflect weighted combinations or non-linear transformations of the dimensions in the original space.

Dimensionality reduction

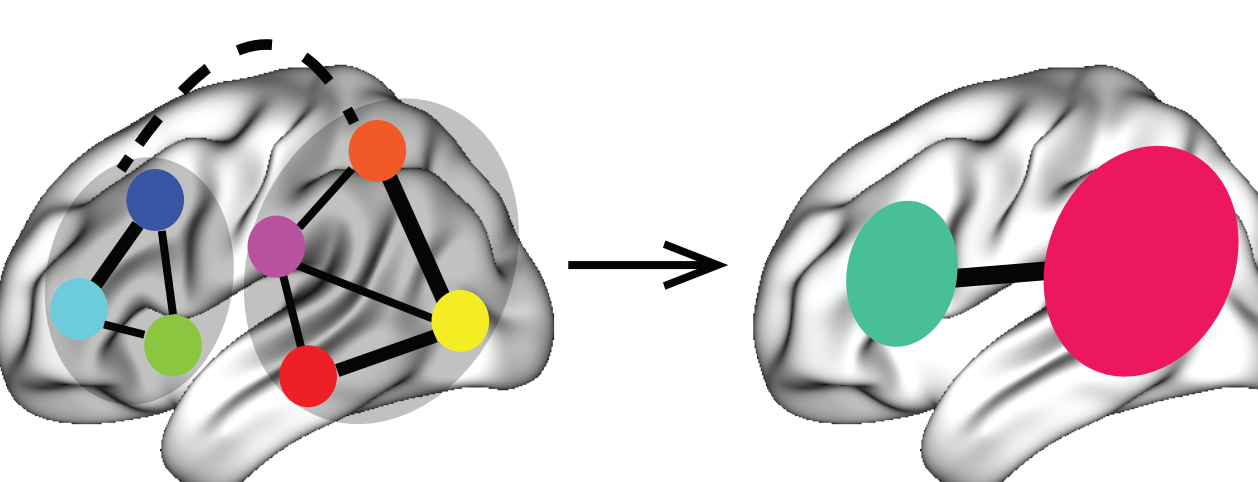


Fig 5. Dimensionality reduction techniques transform high-dimensional data into a lower-dimensional representation while preserving as much structure as possible. This simplifies analysis and reduces noise.

PCA

Identifies new axes (principal components) that capture the directions of greatest variation in the data. The first few components provide a compact summary of the dataset while retaining most of its information.

Incremental PCA

A version of PCA designed for very large datasets. It computes principal components in smaller batches, reducing memory use while producing nearly identical results to standard PCA.

TruncatedSVD

Decomposes the data matrix into a small number of singular vectors and values. Like PCA, it captures the strongest patterns of variation, but it works efficiently on sparse or high-dimensional data.

Factor Analysis

Models observed variables as combinations of a few hidden factors that explain their shared patterns. FA is useful for uncovering latent sources that drive correlations in the data.

Effect of Window Profile

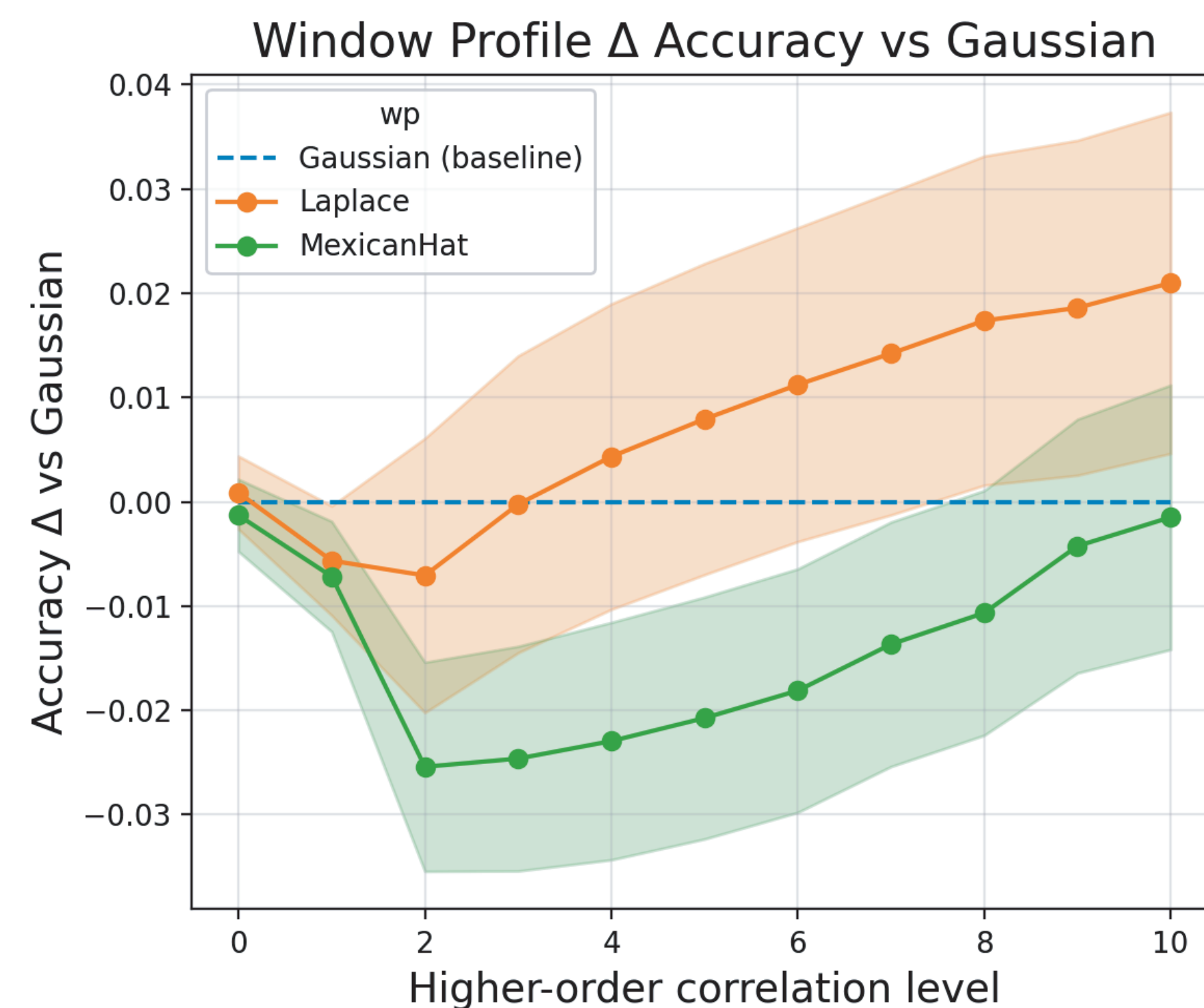


Fig 7. The window profile graph shows a significant improvement of the Laplace kernel over the Mexican-Hat profile, while both Laplace and Mexican-hat did not differ significantly from the Gaussian baseline, suggesting that smoother kernels generally maintain comparable decoding performance.

Effect of Temporal Kernel Width

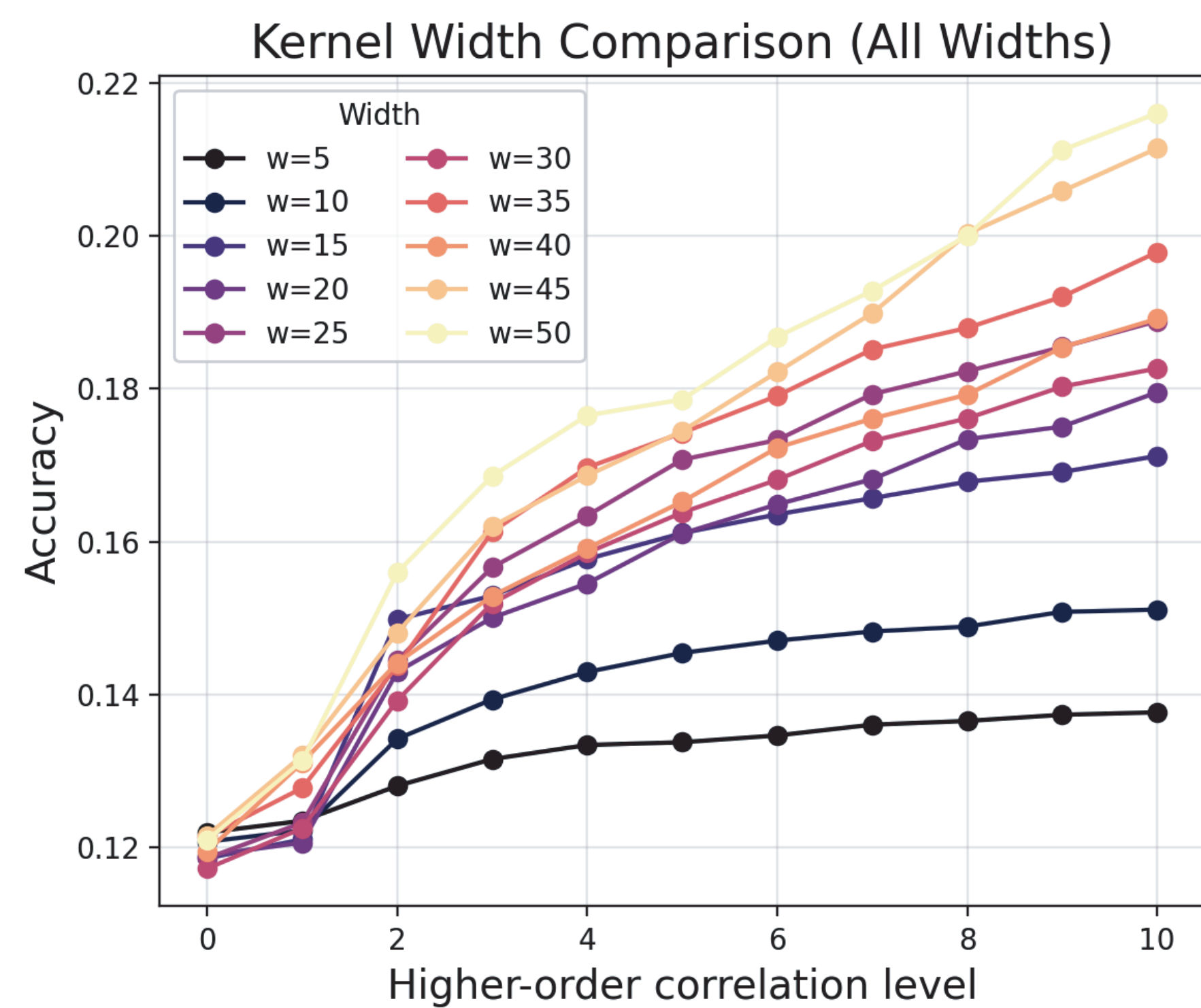


Fig 8. The temporal kernel width graph shows that decoding accuracy increases steadily with broader kernels, with a statistically significant improvement for larger widths (45-50) compared to smaller widths (5-10). This indicates that extended temporal integration enhances the detection of coherent neural patterns.

Results, Cont.

Integrated Parameter Effects

Infection and Plateau Across Methods, Conditions, and Kernel Width

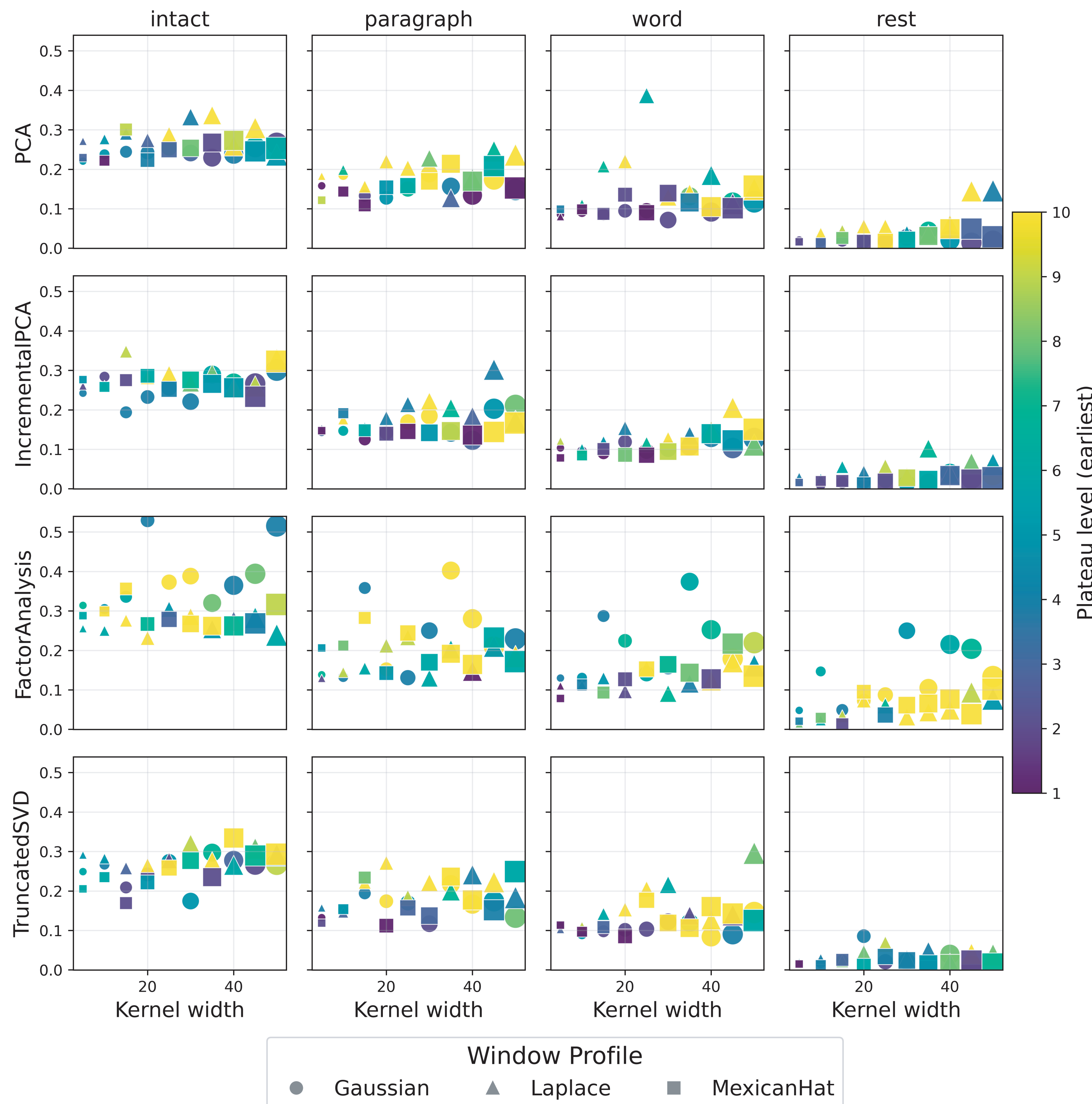
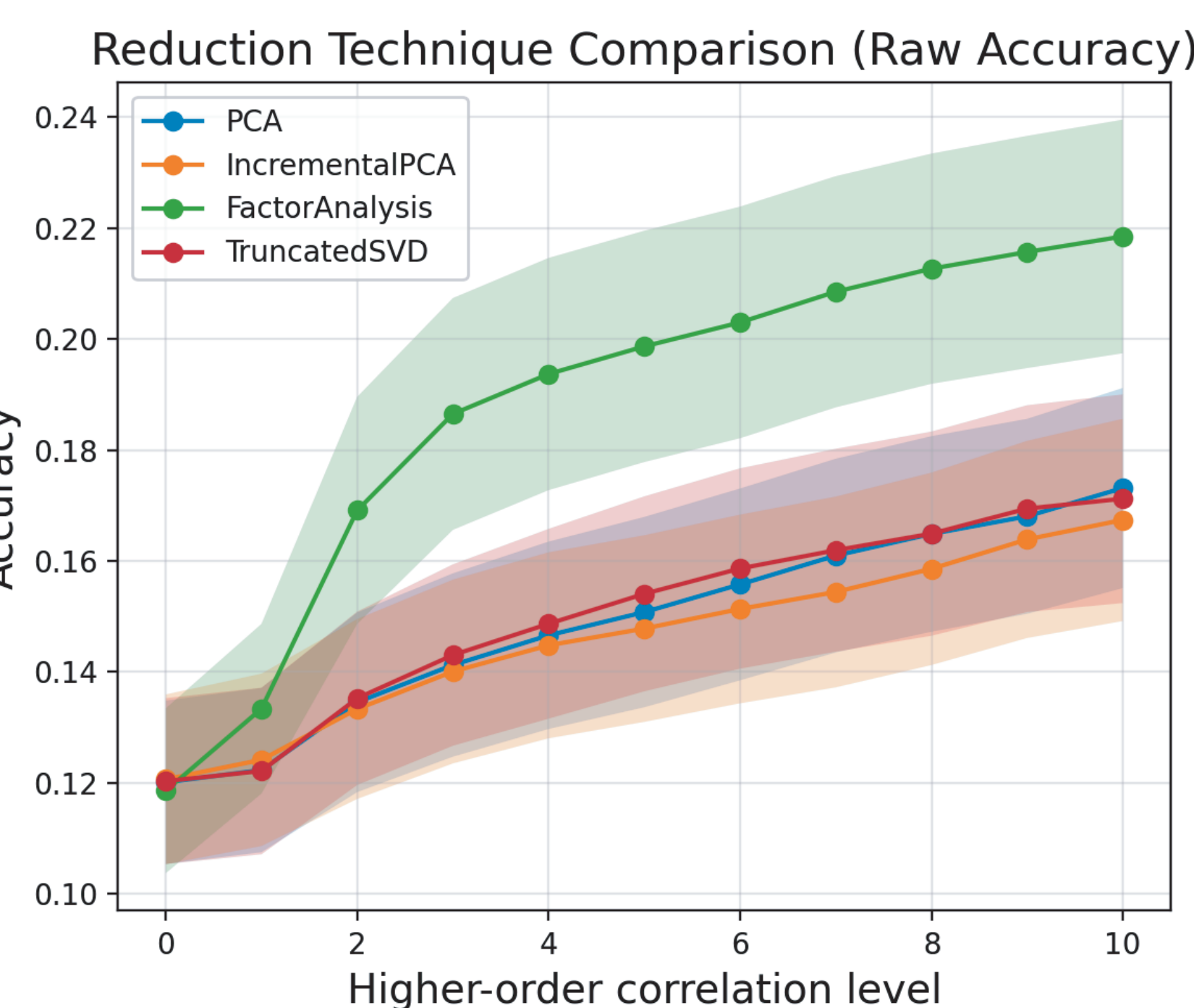


Fig 9. To estimate where performance "levels off," each accuracy curve was first smoothed with a small 3-point moving-average window. The inflection point was defined as where the second derivative changed sign. This marks the point at which increasing the correlation level no longer provides meaningful gains.

The Gaussian + FactorAnalysis combination stands out as an outlier, achieving higher accuracy than neighboring configurations and plateauing at relatively low higher-order correlations. This suggests that FA might be capturing some underlying "hidden" temporal components that align really well with Gaussian weighting, which drives the increased correlations concentrated around the 2nd - 3rd higher-order levels.

Results

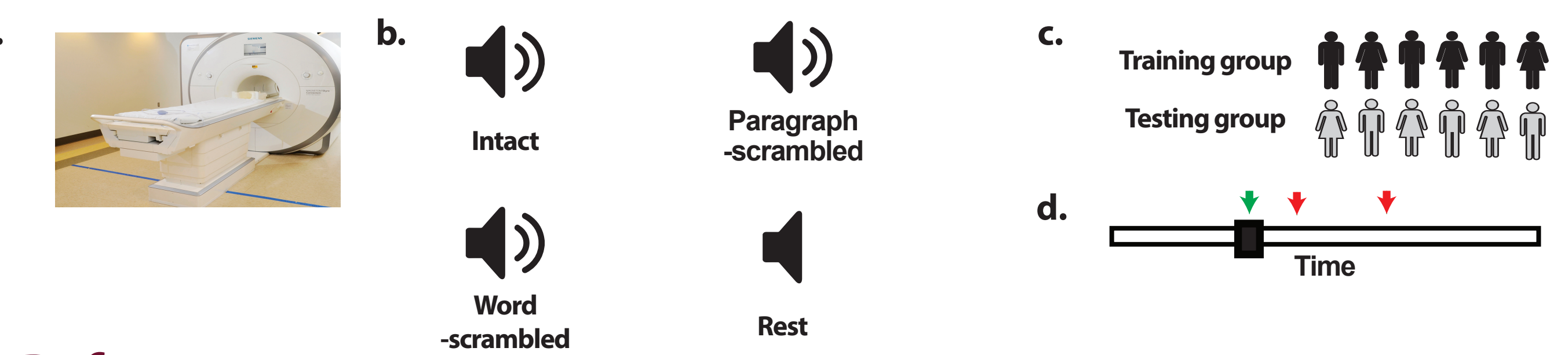


Effect of Dimensionality Reduction Technique

Fig 6. The dimensionality reduction graph shows that Factor Analysis consistently outperforms PCA, IncrementalPCA, and TruncatedSVD across higher order correlation levels. The improvement becomes statistically significant beginning at the second order correlation.

Methods

Fig. 9 a. Participants lay in the scanner for ~10 minutes while functional data were collected. b. Participants were randomly assigned to 1 of 4 experimental conditions (story told either intact, paragraph-scrambled, word-scrambled, or eyes open rest). The experimental conditions varied systematically in cognitive "richness" with audio recording of the story Pie Man by Jim O'Grady. We applied HTFA (Manning et al., 2018) to obtain 700 node activities for every participant. c. We randomly assigned participants in each condition to two groups. We computed dynamic inter-subject functional connectivity (ISFC) for up to 10th order for each group. We then computed optimal weighted-mixture of the ordered correlations for each group. d. We performed timepoint decoding by compared the in-fold groups' activity patterns (using Pearson correlations) to estimate the story times each corresponding out-fold patterns.



References

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